



Convolutional autoencoder-based SOH estimation of lithium-ion batteries using electrochemical impedance spectroscopy

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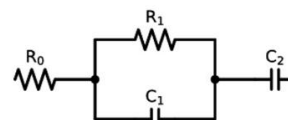
Introduction

Crucial Role of Li-ion Batteries:

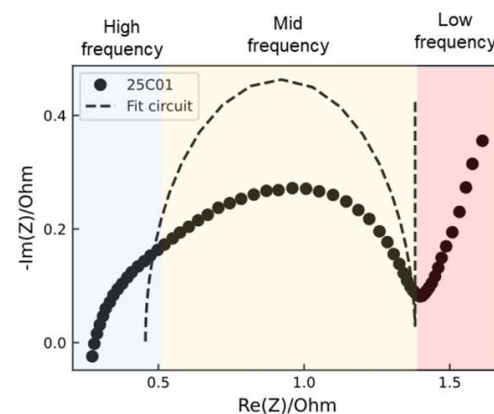
- Li-ion batteries are central to EVs and consumer electronics, requiring accurate **State-of-Health (SOH)** monitoring for safety and longevity.

Limitations of Traditional Methods?

- Equivalent Circuit Models (ECMs):** Rely on constructing physical circuit models which require complex parameter identification.
- Manual Feature Extraction:** Methods like Gray analysis or SVD are time-consuming and prone to human error.
- The Gap:** There is a need for an **automatic method** that doesn't rely on pre-defined physical models.



(a) Simplified equivalent circuit



(b) Nyquist plot



Introduction

Why use EIS Data?:

- EIS (Electrochemical Impedance Spectroscopy) sees "inside" the battery .
- As shown in Fig. 2: The graph changes clearly when SOH drops (from 100% to 80%) .
- This makes it great for tracking battery health.

Our Solution: Deep Learning?

- We use a model called CAE (Convolutional Autoencoder).
- It learns features automatically from EIS data.
- **Benefit:** No need for manual calculation or physical models.

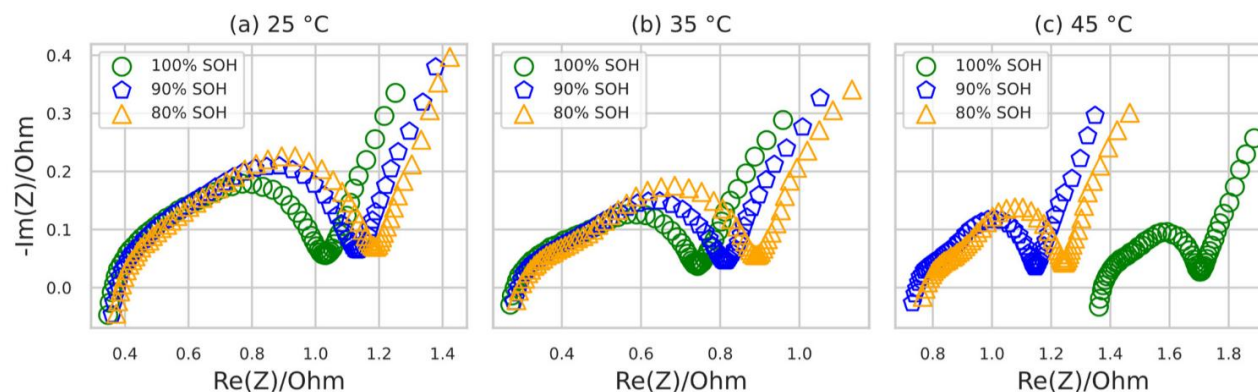


Fig. 2. Nyquist plots for 100, 90, and 80 % SOH for three cells at (a) 25 (25C01), (b) 35 (35C01), and (c) 45 °C (45C01). The real part of the impedance is plotted on the x-axis, and the imaginary part is plotted on the y-axis.



Objectives & Contributions

Main Objective:

- To build a Deep Learning model (**CAE-DNN**) for SOH estimation using only EIS data.

Our Key Innovations?

- New Data Format:** We transform raw EIS data into **2D Images** to work better with AI.
- Automatic Learning:** We use an **Autoencoder** to extract features automatically. No need for manual calculation or labeling.

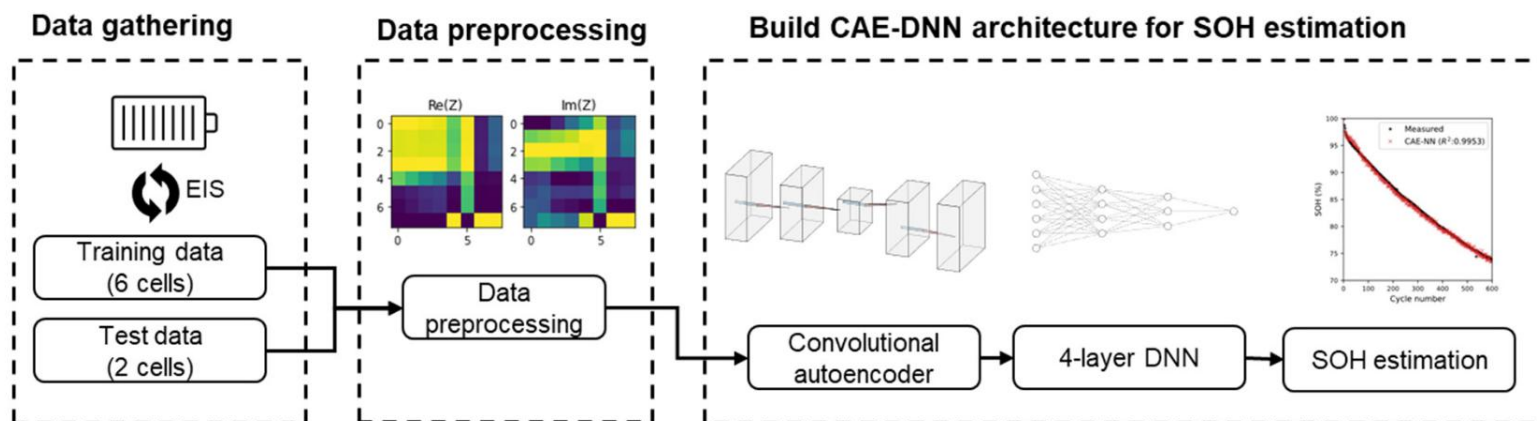


Fig. 3. Proposed methodology overview for SOH prediction using CAE and DNN.



Data Preprocessing

Transforming Data into "Images"

Raw Data Input:

- We measure impedance (Z) at 60 different frequencies.
- Total 120 values (Real and Imaginary parts).

Key Steps

- **Normalize:** Scale the values to make them easy to learn.
- **Add Statistics:** We add 4 values (Min, Max, Mean, Std) to give more information.
- **Reshape to 2D:** We turn the data into an 8x8 matrix.

Result

- The data becomes a 2D Image with 2 channels (Real & Imaginary).

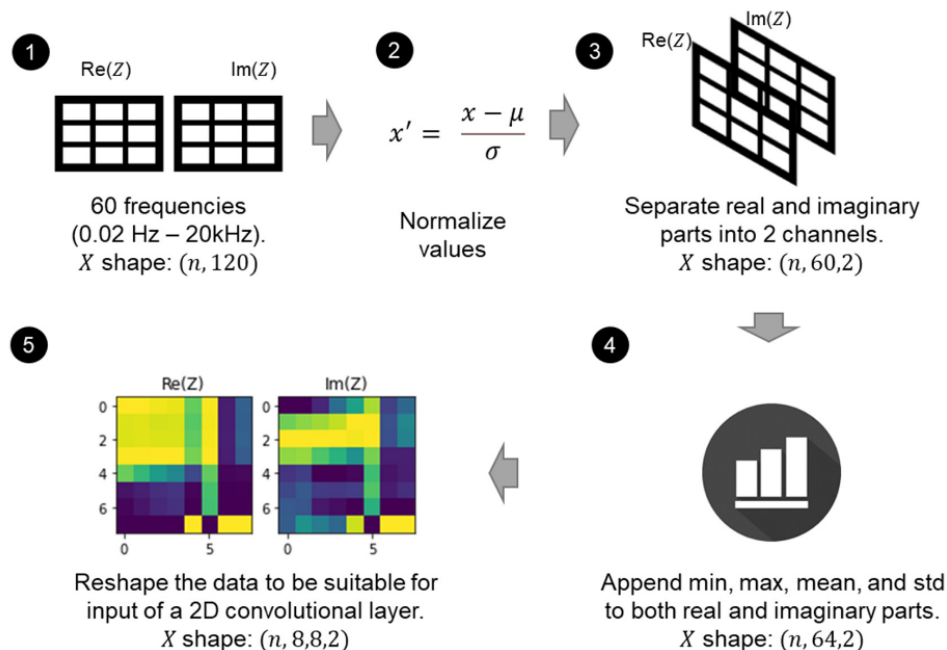


Fig. 4. Preprocessing steps applied to ensure EIS data suitability for use in the proposed deep learning network.



Model Architecture (CAE-DNN)

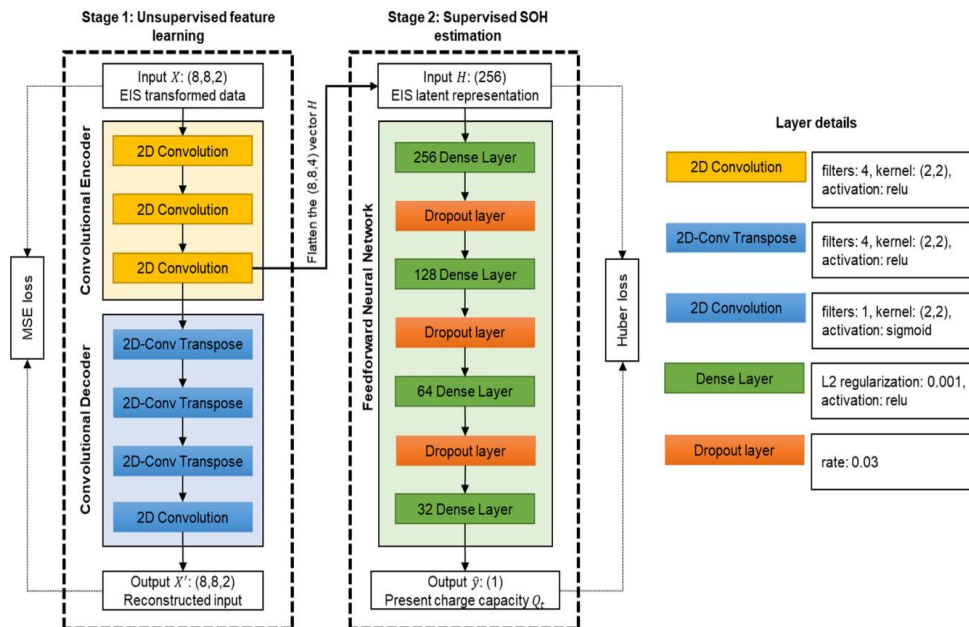
A Two-Stage Deep Learning Model

Stage 1: Unsupervised Feature Learning (CAE):

- We use a **Convolutional Autoencoder (CAE)**.
- **Input:** The 8×8 image from the previous step.
- **Process:** The model learns to compress the image into a **Latent Vector** (size 256) automatically.
- **Goal:** To extract the best features and remove noise.

Stage 2: Supervised SOH Estimation (DNN):

- We use a **Deep Neural Network (DNN)**.
- **Input:** The **Latent Vector** from Stage 1.
- **Structure:** It has 4 hidden layers (256 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32).
- **Output:** The model predicts the **Charge Capacity (Q_t)** to find the SOH.





Experimental Setup

The Batteries:

- We used 8 Li-ion coin cells (Eunicell LR2032).

Test Conditions:

- Tested at 3 temperatures: 25°C, 35°C, and 45°C.

Data Splitting :

- **Training:** 6 cells (mostly at 25°C.)
- **Testing:** 2 cells (at 35°C and 45°C.)
- **Goal:** To check if the model works well on **new temperatures**.

Comparison Models:

- We compared our model against 5 methods: **GPR**, **RF**, **SVR**, **FNN**, and **CNN**.

Table 3

The hyperparameters of the baseline models used in the experiments.

Algorithm	Parameters
GPR	Zero mean function, squared exponential covariance function, Gaussian likelihood function
RF	$estimators = 100$, $\alpha = 0.001$
SVR	$kernel = RBF$, $C = 1000$, $\epsilon = 0.1$
FNN	$neurons = 128$, $L2 = 0.001$, $g = ReLU$
CNN	$Conv2D$ (16 filters (3, 3)), $Conv2D$ (32 filters (3, 3)), DNN (320, 160, 80 neurons), $L2 = 0.001$, $g = ReLU$



Result Quantitative

The Winner:

- The CAE-DNN model performed the best among all 6 models.

Test at 35°C:

- CAE-DNN got the highest accuracy ($R^2 = 0.9657$).

Test at 45°C (Critical Result):

- Standard models failed. For example, normal CNN dropped to $R^2 = 0.35$.
- CAE-DNN remained very accurate ($R^2 = 0.97$).
- *Conclusion:* CAE-DNN is robust even at high temperatures.

Table 4

Performance evaluation of experiments conducted for the test cells of all models. The best measure for each temperature is presented in bold and underlined.

Cell	Temperature	Prediction Model	RMSE	MAPE	R^2
35C02	35 ° C	GPR	0.0149	<u>0.0127</u>	0.9545
		RF	0.0650	0.0807	0.1304
		SVR	0.0266	0.0314	0.8544
		FNN	0.0662	0.0689	0.0962
		CNN	0.0172	0.0131	0.9387
		CAE-DNN	<u>0.0129</u>	0.0128	<u>0.9657</u>
45C02	45 ° C	GPR	0.0279	0.0290	0.8288
		RF	0.0425	0.0499	0.6028
		SVR	0.0382	0.0411	0.6796
		FNN	0.0609	0.0632	0.1858
		CNN	0.0541	0.0628	0.3565
		CAE-DNN	<u>0.0112</u>	<u>0.0118</u>	<u>0.9723</u>



Result Visual

Test at 45°C

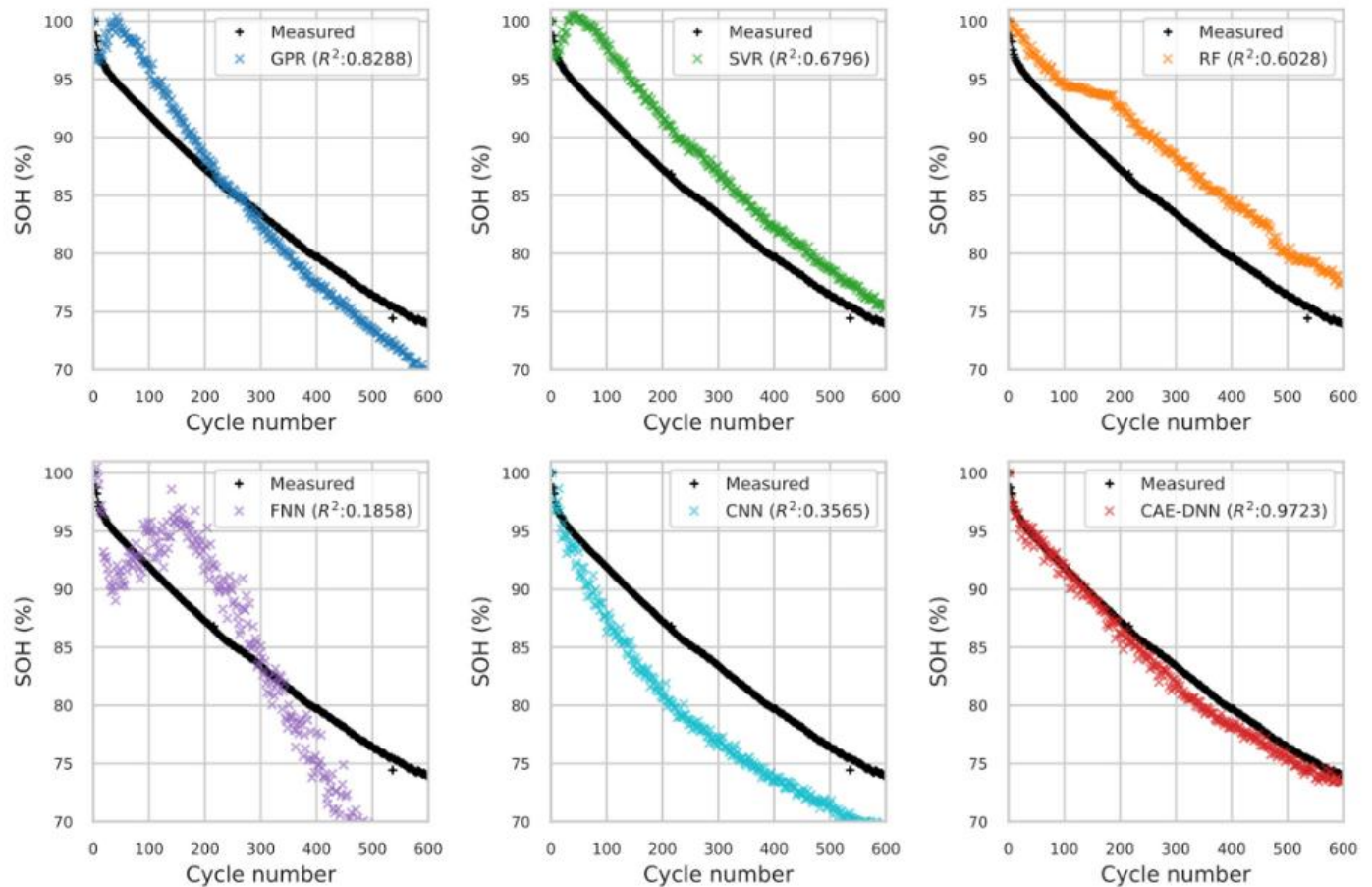


Fig. 8. Estimated (blue, green, orange, purple, cyan, and red curves) and measured (black curve) SOH as a function of the number of cycles for the 45C02 test cell. The predicted values along with the R^2 measure are presented for GPR, SVR, RF, FNN, CNN and the proposed CAE-DNN model. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



Discussion and Conclusion

Ablation Study (Testing Parameters):

- We tested different model settings to find the best one.
- **Filters:** Adding more filters did not help. **4 filters** were enough to learn features.
- **Kernels:** Bigger kernels (3x3) were worse because our input image (8×8) is small.

Optimal Settings (As seen in Table 5):

- The best result came from using **4 filters** and a **dropout rate of 0.03**.
- Changing these values increased the error

Why is CAE better than CNN?:

- **Unsupervised Learning:** The Autoencoder extracts features automatically without labels
- **Denoising:** It helps remove **noise** from the EIS data, making the model more robust than a standard CNN

Table 5

Ablation studies conducted on main CAE-DNN hyperparameters. The results presented are on the test cells. The best measure for each temperature is highlighted in bold and underlined.

Layer	Ablated hyperparameter	Value	35C02			45C02		
			RMSE	MAPE	R^2	RMSE	MAPE	R^2
Proposed model ($f = 4, k = (2,2), r = 0.03$)			0.0129	0.0128	0.9657	0.0112	0.0118	0.9723
2D Convolution and 2D-Conv Transpose layers (in CAE)	filters	$f = 2$	0.0647	0.0761	0.1377	0.0519	0.0599	0.4090
		$f = 8$	0.0527	0.0671	0.4270	0.0296	0.0322	0.8070
		$f = 16$	0.0303	0.0338	0.8105	0.0564	0.0666	0.3018
Dropout layer (in DNN)	kernel	$k = (3,3)$	0.0322	0.0342	0.7864	0.0212	0.0204	0.9008
	dropout rate	$r = 0.01$	0.0268	0.0267	0.8523	0.0344	0.0404	0.7396
		$r = 0.05$	0.0565	0.0699	0.3429	0.0405	0.0458	0.6394
		$r = 0.1$	0.0642	0.0729	0.1511	0.0294	0.0329	0.8096



Discussion and Conclusion

Successful Method:

- We successfully created a Deep Learning model (**CAE-DNN**) to estimate battery health using EIS data.

Key Innovation:

- Transforming EIS data into **2D images** allows the model to learn hidden features automatically.

Performance:

- The model works **very well** even at high temperatures (up to 45°), beating standard methods like GPR and CNN.

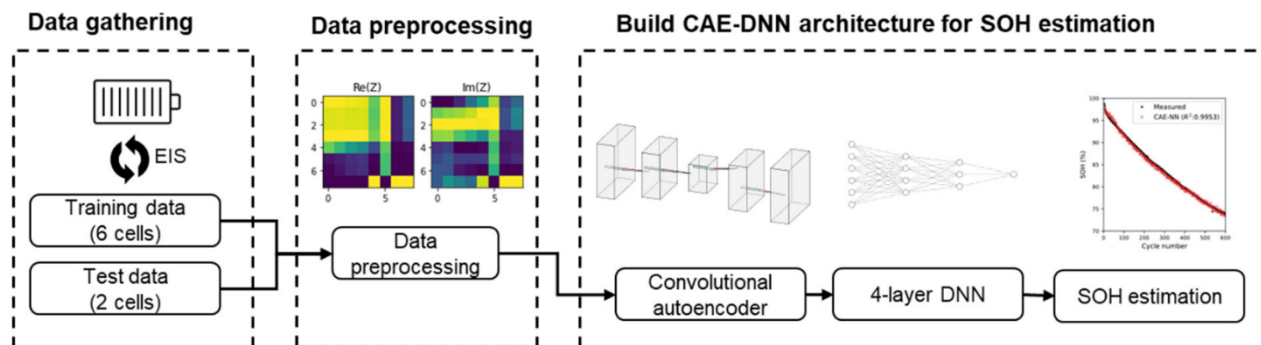


Fig. 3. Proposed methodology overview for SOH prediction using CAE and DNN.



Critical Review & Future Work

Strengths (What is Good):

- **Smart Idea:** Converting EIS data into **2D images** is very clever. It makes it easy to use advanced AI models (**CNNs**).
- **Robustness:** The model works well even in **hot environments** (45°C), which is great for safety.

Limitations (What is Missing):

- **Full Charge Only:** The paper only tested batteries at **100% SOC** (Fully Charged)
- **Real World Issue:** In real life (EVs), we **cannot always wait** for a full charge to check health.

Future Work (What to do next):

- **Partial Charging:** We need to test the model with data from **Partial Charging** (e.g., at 50% battery level).
- **New Models:** Try using newer AI models such like **Transformers** to see if they are better.



Thank you